

# WASP-50b

R-1027

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-50\_161210 · created 2026-07-30T05:48:58+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                               |
|---------------------------------------------------|-------------------------------|
| Mid-Transit Time (Tmid)                           | 2457732.714 +/- 0.035 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.171 +/- 0.049               |
| Transit depth (Rp/Rs)^2                           | 2.93 +/- 1.69 [%]             |
| Orbital Inclination (inc)                         | 83.51 +/- 2.87                |
| Airmass coefficient 1 (a1)                        | 0.948 +/- 0.073               |
| Airmass coefficient 2 (a2)                        | 0.03 +/- 0.087                |
| Scatter in the residuals of the lightcurve fit is | 7.66 %                        |
| Best Comparison Star                              | #1 - [534, 20]                |
| Optimal Method                                    | PSF photometry                |
| Transit Duration (day)                            | 0.056 +/- 0.033               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R* fit vs published    | 0.171 ± 0.049 vs 0.1404 (deviation 0.58σ) |
| Duration fit vs published | 0.056 d vs 0.0754 d (deviation 0.55σ)     |
| Mid-time O–C              | 54.8 min                                  |
| Depth SNR                 | 3.2                                       |
| Residual scatter          | 7.66 %                                    |

## Light curve

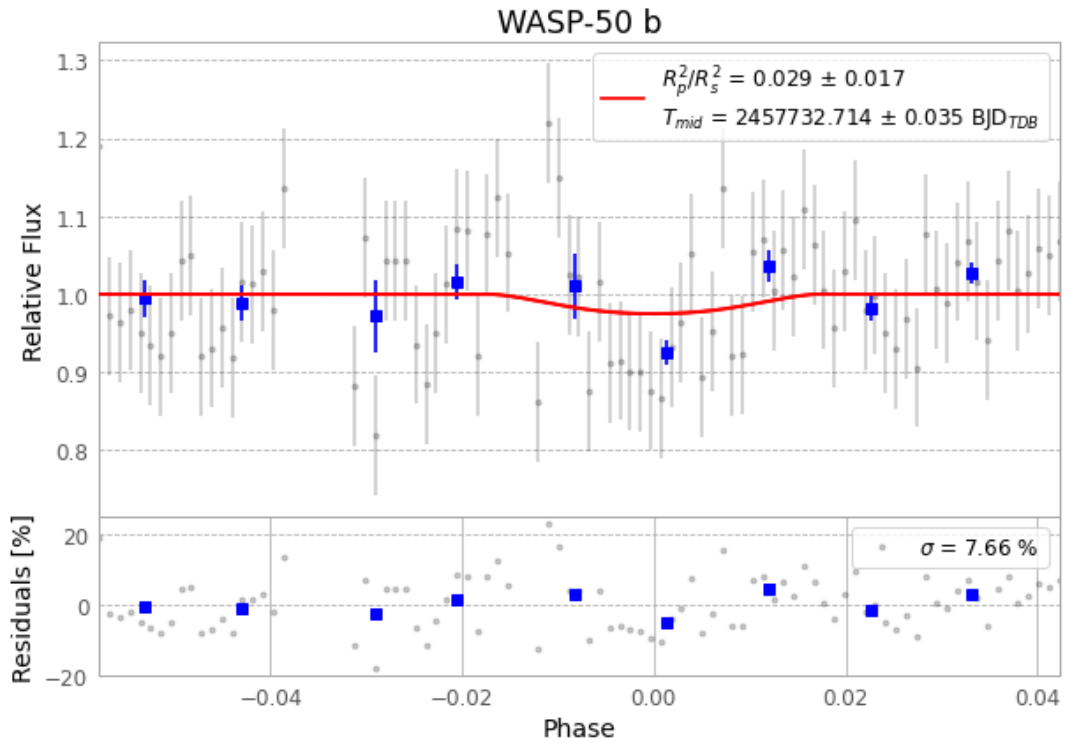


Figure 1 — FinalLightCurve\_WASP-50 b\_2016-12-09.png (SHA-256 e55ab872699f99ca...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [311, 106], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 06e6806d4a6d01c5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 33c71d4

## Data provenance

95 FITS frames (62 MB), WASP-50161210021812.FITS → WASP-50161210070012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-161210.log (1542abd30c65...), FinalLightCurve\_WASP-50 b\_2016-12-09.png (e55ab872699f...), FinalLightCurve\_WASP-50 b\_2016-12-09.csv (6f1c75377c65...)

## Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 05:48Z.